



OUR AIM IS TO PROMOTE OUTSANDING DATA TRANSFORMATION AND
ENGINEERING, THUS FACILITATING THE GENERATION OF VALUABLE INSIGHTS,
AND OPENING A WORLD OF LIMITLESS POSSIBILITIES WITH DATA LEAD
DECISION MAKING

Biliv-Augmenta Employee Attrition Analysis Project

Phase 1: Business Objective

Employee attrition, the loss of employees due to resignation, retirement, or other reasons, poses a significant challenge for companies, especially when vacancies remain unfilled. High attrition rates can lead to increased costs, reduced productivity, and the loss of institutional knowledge. To maintain a competitive edge, companies must not only attract skilled talent but also retain them over the long term.

As a Data Analyst at Biliv-Augmenta, you have been assigned the task of analyzing the company's employee attrition trends to identify the key factors contributing to turnover. Your goal is to examine the provided employee data, uncover patterns, and generate actionable insights that will help the organization strengthen its employee retention strategies. Ultimately, your analysis will inform the development of more effective programs aimed at reducing attrition and improving employee satisfaction.

To support this analysis, leverage Microsoft Power BI to perform exploratory data analysis, visualize the data, identify trends, and deliver comprehensive insights to stakeholders. **Note:** The purpose of this analysis will help the organization determine what action they will need to take to retain more employees.

Phase-2: Potential Analyses

With this dataset, we can explore the following questions:

1.

-  What are the top factors contributing to employee attrition at Biliv-Augmenta?
-  Are there specific job roles, departments, or demographics more prone to attrition?
-  How does job satisfaction, work-life balance, or compensation affect turnover?
-  Can patterns be identified based on tenure, promotion history, and involvement in training?

Please note: You are not required to answer all of the questions listed below. Instead, use them as a guide to steer the Exploratory Data Analysis (EDA) process, focusing on the areas that provide the most meaningful insights for your analysis.

Additional questions we can explore using the Biliv-EmpDt dataset to deepen the analysis of employee attrition at Biliv-Augmenta:

2. Attrition by Performance:

- a. Are lower performance ratings (both self and manager ratings) correlated with higher attrition rates?
- b. Do employees who have fewer training opportunities or do not engage in training have a higher likelihood of leaving?

3. Attrition by Satisfaction Levels:

- a. How does the level of job satisfaction, environment satisfaction, and work-life balance impact the attrition rates across different departments and job roles?
- b. Is there a significant difference in attrition rates between employees who are "Very Satisfied" versus those who are "Dissatisfied"?

4. Impact of Career Progression:

- a. Are employees who have been in their current role for many years without promotion more likely to leave?
- b. How does the number of years with the current manager affect attrition trends?
- c. Does a lack of career advancement opportunities (as indicated by YearsInCurrentRole or YearsSinceLastPromotion) contribute significantly to employee turnover?

5. Compensation and Benefits:

- a. Do employees with higher salary hikes or stock options tend to stay longer in the company?
- b. How does monthly income correlate with attrition? Are higher earners more likely to stay?

6. Demographic-Based Insights:

- a. Are younger employees (e.g., those in the 20-29 age range) more likely to leave compared to older employees?

- b. Is there a noticeable gender disparity in attrition rates? Do male, female, and non-binary employees exhibit different retention trends?

7. Work-Life Balance and OverTime:

- a. Is there a significant correlation between working overtime and higher attrition rates?
- b. How does dissatisfaction with work-life balance drive attrition in different departments and job roles?

8. Attrition Trends Over Time:

- a. Has the attrition rate changed over the years? If so, what factors may be driving these changes (e.g., economic conditions, company policies)?
- b. Can any seasonal patterns be identified in the attrition data? For example, is there a particular month or quarter when employees tend to leave?

9. Tenure and Experience:

- a. What is the relationship between the total number of years an employee has worked (TotalWorkingYears) and the likelihood of leaving the company?
- b. Do employees who have worked at multiple companies (NumCompaniesWorked) before joining Biliv-Augmenta have a higher attrition rate?

10. Geographic Considerations:

- a. Do employees who live farther from the office (DistanceFromHome) exhibit higher attrition rates due to commute issues?
- b. Are there regional or state-based trends that affect retention, especially if employees from certain locations face challenges in work-life balance or job satisfaction?

11. Impact of Marital Status:

- a. Does marital status (single, married, divorced) have any correlation with retention or attrition rates?
- b. Are there any differences in work-life balance satisfaction between single and married employees that may contribute to turnover?

These questions can guide a more in-depth exploration of the factors influencing employee attrition and provide valuable insights for Biliv-Augmenta's retention strategy.